

Fortress Notes, in plain language

A local-first notes app. It treats each paragraph, heading, image, and other piece as a block that can be edited and organized.

What can I do?

Write and format

- Use bold, italic, links, code, highlights, headings, lists, quotes, and callouts.
- Add images and files. A math block is currently a placeholder.
- Copy and paste formatted text through the editor's HTML clipboard.

Organize and connect

- Group documents in folders and reopen recent work.
- Reorder and nest blocks; add tags and key/value details.
- Link documents and blocks, then follow backlinks or a copied block link.

Find and understand

- Search inside individual blocks and filter results.
- See related blocks and why they matched.
- Image OCR text can be searched with the block that owns the image.

Keep and move your work

- Changes save as transactions with revision checks and persistent undo/redo.
- Unsent edits are queued in browser storage and replay after reload.
- Import preview, Markdown export, and database backup/restore support portability.

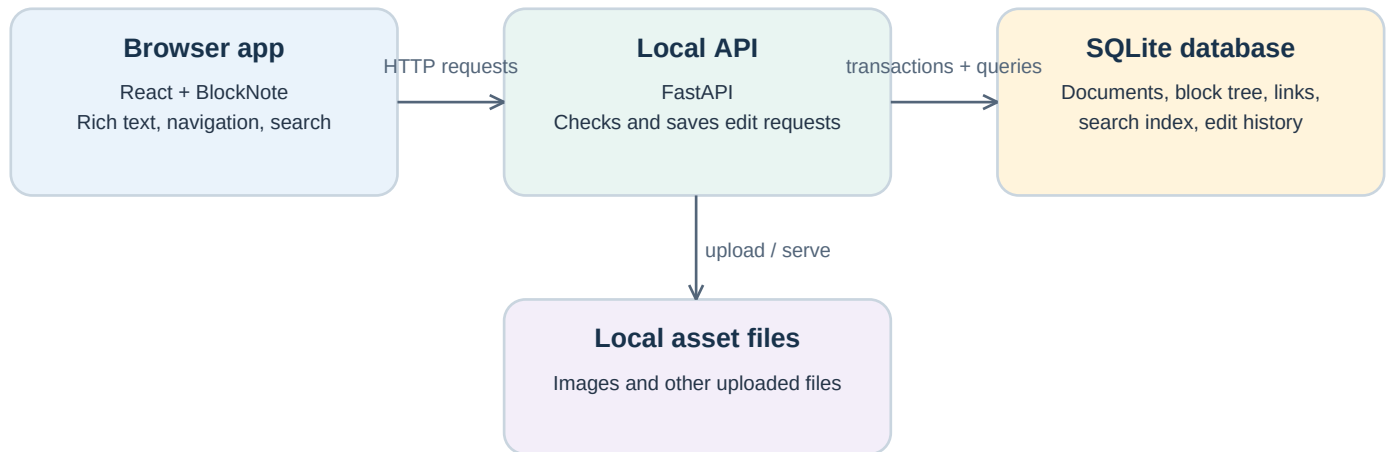
AI features

Review a current block, selected blocks, the whole document, or directly linked blocks. Findings show the saved block ID and an exact quote. Users can accept or reject findings and keep those decisions across reloads. Automatic edits can update exact text split across runs when all runs use the same formatting. Mixed formatting, links, stale text, and non-text content remain review-only. AI checks send the selected note or block text to the configured OpenRouter service.

What “rich text” means here: People edit with document-style controls, including Ctrl/Command+B for bold and Ctrl/Command+I for italic, and copy/paste formatted HTML. Markdown remains a compatibility/storage format; people do not need to type Markdown to format a note. Browser clipboard checks confirm that copied local images include their bytes. Paste into Word and Docs was not checked at the user's request.

How the pieces fit together

The browser handles editing. The local API checks and saves changes. SQLite is the main record of documents and blocks; uploaded media stays in local files.



The main idea

A document is a tree of blocks. A paragraph, heading, image, or list item has its own stable ID. Blocks can contain child blocks. This makes it possible to link to a specific block, search at that level, and undo a saved transaction.

Edits are saved in groups. The editor sends a transaction with the revision it started from. The API checks the revision and writes the changes together. SQLite also keeps history for undo/redo. Pending browser edits can be replayed safely after reload.

Rich formatting is stored with the block. BlockNote handles the editing experience and rich HTML copy/paste. Markdown is used for compatibility import/export and older endpoints. Image/file bytes are separate local assets; the block stores the reference to them.

Where code lives

Path	Purpose
frontend/src/features/notes/	The note editor, sidebar, search panels, links, and other screen features.
backend/app/	The local API, SQLite block store, search, assets, import/export, and AI adapters.
backend/tests/	Focused tests for storage rules, APIs, import/export, search, and recovery.
notes/	The configured local notes and uploaded assets; SQLite database is stored here too.

Verified: keyboard formatting, accessible editor and toolbar names, repeatable browser editing, rich HTML clipboard, embedded local-image bytes, reload recovery, and undo/redo. Word/Docs paste was not checked at the user's request.